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PROFESSIONAL SUMMARY

AI Engineer with 2 years of hands-on experience developing and evaluating AI solutions across Computer Vision, Machine Learning, Deep Learning, and NLP. Project experience spans instance segmentation, object detection, image classification, regression, and statistical data analysis. Skilled in model training and fine-tuning, dataset curation, experimental design, ablation studies, quantitative benchmarking, and error analysis, with a strong focus on research-driven development and rigorous model evaluation.

EDUCATION

Damascus University - Faculty of Informatics Engineering
B.Sc. in Artificial Intelligence (2026)

KEY PROJECTS

Vehicle Inspection System (VIS) - Multi-View Vehicle Inspection | *Graduation Project, 2026*

YOLO26L-Seg • SAM • PyTorch • OpenCV • Instance Segmentation • Object Detection • Ensemble Voting

- Built an end-to-end multi-view vehicle inspection system combining 20-class vehicle-part instance segmentation with 8-class external-damage detection and segmentation.
- Extended a SAV-based segmentation methodology with SAM-assisted mask refinement, confidence-aware selection, and hybrid refinement for challenging vehicle regions.
- Benchmarked the proposed approach against Mask R-CNN, GCNet, PANet, CBNet, and HTC, achieving 68.5 AP50-95, 90.9 AP50, and 73.4 AP75 - 13.1 AP points above the strongest evaluated baseline.
- Developed the damage pipeline using specialized YOLO detectors, instance segmentation, multi-scale processing, Boundary Loss, and ensemble voting for small and irregular damage, achieving 72.0% F1 and 80.4% recall.

Acne Severity Classification & Lesion Analysis | *2025*

ResNet50 • MediaPipe • OpenCV • Adaptive-LoG • PyTorch • Transfer Learning

- Developed a ResNet50-based acne severity classification pipeline using face-aware preprocessing, RGB and Adaptive-LoG representations, transfer learning, and lesion-focused feature extraction.
- Engineered a face-aware preprocessing pipeline using MediaPipe Face Mesh, LAB/HSV analysis, CLAHE, adaptive thresholding, LoG filtering, and morphological processing to emphasize acne-related skin regions.
- Fine-tuned an ImageNet-pretrained ResNet50 with partial layer freezing and class-weighted training, evaluating multiple preprocessing and model configurations.

Arabic Punctuation Restoration | *2025*

AraBERTv2 • PyTorch • Hugging Face Transformers • Token Classification • NLP

- Built an AraBERTv2 token-classification system for Arabic punctuation restoration, implementing text normalization, Transformer tokenization, and word-to-subword label alignment.
- Addressed severe class imbalance using weighted cross-entropy loss and rare-label enrichment, with checkpointing, per-class evaluation, and confusion-matrix analysis.

Car Price Prediction | *2025*

Python • Scikit-learn • Linear Regression • Feature Engineering • Cross-Validation

- Prepared and cleaned the vehicle-price dataset using EDA, IQR-based outlier clipping, categorical encoding, and train-only feature scaling, then engineered predictive features including power-to-weight, average MPG, and power-to-size.
- Built a supervised Linear Regression pipeline with Scikit-learn and evaluated it using train-test validation and 5-fold cross-validation, with coefficient and error analysis to assess model behavior and generalization.

Uganda RCT Data Analysis - A Mother's Voice | *Coursework*

Pandas • NumPy • SciPy • Matplotlib • Data Cleaning • EDA • Statistical Analysis

- Cleaned and linked multi-level randomized-trial survey data covering 5,512 families and 22,308 child records, including missing-value normalization, structural-column removal, ID-based linkage, and treatment/district summaries.
- Conducted exploratory and statistical analysis using Spearman correlation tests to investigate relationships involving household decision-making, maternal education, income, and nutrition-related change variables.

Hashtag Social Media Application | 2025

Flutter • Dart • BLoC • Dio / REST APIs • Video Trimmer

- Developed an Arabic-first Flutter social application using BLoC, repository-based data access, GetIt dependency injection, and GoRouter, supporting authentication, profiles, search, follow interactions, and short-media workflows.
- Integrated Dio REST APIs, secure token storage, media upload workflows, and on-device video trimming, together with like, save, report, and profile-content features.

TECHNICAL SKILLS

- **Languages:** Python, Dart
- **ML / DL & CV Frameworks:** PyTorch, Scikit-learn, OpenCV, MediaPipe, Hugging Face Transformers
- **Models & Architectures:** YOLO / YOLO-Seg, SAM, ResNet50, AraBERTv2, CNN Architectures
- **Training & Modeling Methods:** Transfer Learning, Fine-Tuning, Feature Fusion, Ensemble Methods, Boundary Loss, Class-Imbalance Handling
- **Data & Experimentation:** NumPy, Pandas, SciPy, Matplotlib, Seaborn, Dataset Curation, Feature Engineering, Cross-Validation, Ablation Studies, Model Evaluation, Error Analysis, Statistical Analysis
- **Backend & Mobile:** FastAPI, REST APIs, Flutter, BLoC, Dio
- **Tools & Platforms:** Git/GitHub, Jupyter Notebook, Google Colab, Kaggle, VS Code
- **Core AI Areas:** Computer Vision, Object Detection, Instance Segmentation, Image Classification, Image Processing, Machine Learning, Deep Learning, NLP, Token Classification

LANGUAGES

- Arabic.
- English.